



Department of Computer Engineering

COMPLEX ENGINEERING PROBLEM

(SEMESTER PROJECT)

Electronic Devices and Circuits

INSTRUCTOR: Engr. Muhammad Yasir Amir Khan

COURSE CODE: EEN-224

SEMESTER: 3 (Fall-2025)

CREDIT HOURS: 3 (Theory)

Complex Engineering Problem: Off-Grid Integrated Environmental Monitoring System

Problem Statement:

Design, analyze, and construct a prototype for a compact, off-grid power supply and environmental monitoring system. This system will use a solar panel to charge a 12V battery, which in turn powers both an inverter for AC loads and a suite of integrated sensors for water and rain detection. The system must operate autonomously and manage power efficiently under varying environmental conditions.

Core Project Options:

- **Water Level Indicator & Rain Detection System:** Develop a combined circuit that can both detect the presence of rain and monitor water levels in a tank. The circuit should be reliable and energy-efficient, using minimal power.
 - **Smart Battery Management & Charging:** Design an automatic battery charger that optimizes charging from the solar panel and prevents overcharging and deep discharge.
 - **High-Efficiency 220V AC Inverter:** Construct a high-efficiency inverter that converts the 12V DC power from the battery to a stable 220V AC output to power the remote device, minimizing power loss.
 - **Solar Panel Protection System:** Design a mechanism for the solar panels to automatically adjust their position or fold to protect them from potential damage during extreme weather events.
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CLO Mapping

This project maps directly to the following Course Learning Outcomes (CLOs) from your Electronic Devices & Circuits course:

- CLO-2 (C3): **Apply** the acquired knowledge to solve small scale circuits consisting of semiconductor devices .
This is addressed by the entire project, as students must apply their knowledge of diodes, transistors (BJT/FETs), and other semiconductor devices to build each subsystem (charger, inverter, sensors).
 - CLO-4 (A3): **Demonstrate** necessary tasks required in completion of project work as an individual or as a team member .
The project requires students to be able demonstrate practical tasks such as circuit design, assembly, testing, and debugging.
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CEP Attributes

This project is classified as a Complex Engineering Problem (CEP) because it encompasses several key attributes:

- **Depth of Knowledge Required:** The problem requires a fundamentals-based, first-principles analytical approach. Students will need to use knowledge of semiconductor devices, such as BJT and JFETs, and their DC and AC analysis to design and analyze the amplifier and power management circuits required for the project. This knowledge is at the core of the professional discipline of electronic circuit design.
- **Range of Conflicting Requirements:** The system must be highly efficient, reliable, and durable, while also being cost-effective and portable. This involves trade-offs and compromises, as a high-performance system may not be the most economical or practical solution.
- **Depth of Analysis Required:** The problem has no single obvious solution. Students must engage in abstract thinking to formulate models for power management, efficiency, and system response under varying conditions. The design requires an in-depth analysis of component selection, power consumption, and circuit stability under different environmental stresses.
- **Interdependence:** The project includes multiple component parts that are highly interdependent. The efficiency of the battery charger directly impacts the available power for the inverter and sensors. Similarly, the performance of the solar panel protection system affects the overall power generation and durability of the entire setup.

PROJECT DELIVERABLES

Students are required to submit the following deliverables to be assessed for this Complex Engineering Problem:

1. **Project Report:** A detailed report that includes the problem analysis, design methodology, circuit schematics, bill of materials, and a conclusion on the project's success. The report should also include a section on the challenges faced and how they were overcome.
2. **Circuit Diagrams and Simulations:** Submit a complete set of circuit diagrams created using a suitable software tool (e.g., Eagle, KiCad). This should be accompanied by simulation results (e.g., from Proteus, Multisim) to demonstrate the circuit's functionality and performance.
3. **Physical Prototype:** A fully assembled and functional prototype of the designed system. The prototype should be well-constructed and demonstrate the core functionalities outlined in the chosen project option.
4. **Demonstration and Viva-Voce:** A live demonstration of the prototype's functionality to the instructor, followed by a viva-voce session where students will answer questions about their design choices, analysis, and overall understanding of the project.
5. **Teamwork and Peer Evaluation:** (For teams) A self and peer-evaluation report outlining each team member's specific contributions to the project, demonstrating the 'individual and teamwork' attribute (CLO-4).

COMPLEX ENGINEERING PROBLEM ASSESSMENT RUBRICS:

Rubric	Marks (0-2) (Not Evident)	Marks (2.1-4) (Limited)	Marks (4.1-6) (Developing)	Marks (6.1-8) (Proficient)	Marks (8.1-10) (Exemplary)
Quality of Design	No attempt or completely incorrect approach.	The block diagram and circuit schematics are significantly flawed or incomplete. Component selection is illogical.	Block diagram and schematics have significant errors. Component selection shows basic understanding but with notable limitations. The design may not function as intended.	Block diagram and schematics are mostly correct with minor inaccuracies. Component selection is appropriate. The design applies relevant principles with some effectiveness.	Block diagram and schematics are accurate and well-constructed. Component selection is thorough and justified. The design demonstrates a strong understanding and application of design principles (CLO-2).
Comprehensive Solution	No solution provided or the solution is entirely irrelevant.	The solution addresses very few aspects of the problem and contains major errors. The prototype may not function at all.	The solution addresses some aspects of the problem but contains significant errors or omissions. The prototype may only partially function.	The solution addresses most aspects of the problem with minor errors. The prototype functions as intended but may have minor inefficiencies.	The solution is comprehensive, accurate, and effectively addresses all aspects of the problem. The prototype is highly functional and demonstrates a thorough and insightful understanding of the system (CLO-2).
Individual & Teamwork	Does not contribute to the project or teamwork is non-existent.	Contributions are minimal or work is not integrated with the team. Fails to complete assigned tasks or collaborate effectively.	Contributes to the project but with significant errors or omissions. Teamwork is limited and requires frequent guidance.	Contributes effectively to the project and completes assigned tasks. Teamwork is cooperative, and the individual works well with others.	The individual demonstrates all necessary tasks required in project completion (CLO-4). The student is a proactive and effective team member, showing strong collaboration and initiative.
Accurate Response to Viva Questions	Unable to answer basic questions related to the work.	Provides incorrect or irrelevant answers to most questions. Struggles to explain fundamental concepts related to semiconductor devices and circuits.	Demonstrates a limited understanding of the implemented concepts and provides vague or partially correct answers to questions. Explanations are generally unclear.	Demonstrates a good understanding of the implemented concepts and provides mostly accurate and relevant answers. Explanations are generally clear.	Demonstrates a deep and nuanced understanding of the implemented concepts and provides accurate, insightful, and well-articulated answers to all questions. Can effectively explain and justify all design choices (CLO-2).

COMPLEX ENGINEERING PROBLEM MAPPING WITH CEP CHARACTERISTICS

	Characteristic	Description	Mapping
WP 1	Depth of Knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8	✓
WP 2	Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering or other issues	✓
WP 3	Depth of analysis required	Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models	✓
WP 4	Familiarity of issues	Involve infrequently encountered issues	
WP 5	Extent of applicable codes	Address problems not encompassed by standards and codes of practice for professional engineering	
WP 6	Extent of stakeholder involvement and level of conflicting requirements	Involve diverse groups of stakeholders with widely varying needs.	
WP 7	Interdependence	Address high level problems with many components or sub-problems	✓